



A comparative analysis of deep learning and chemometric approaches for spectral data modeling

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ARTICLE INFO

Dataset link: https://github.com/PClabUC/mo-del_preprocessing_comparison_spectral_analysis

Keywords:

Chemometrics
Spectral modeling
Partial least squares
Deep learning
Model selection

ABSTRACT

Background: This study presents a comprehensive comparison of five different modeling approaches for spectroscopic data analysis. The first approach uses PLS combined with classical chemometric pre-processing (9 models). The second and third approaches use interval PLS (iPLS) with either classical pre-processing or wavelet transforms (28 models). The fourth uses LASSO with wavelet transforms (5 models). Finally, the fifth approach combines CNN with spectral pre-processing (9 models).

Results: In this paper we consider two low dimensional case studies: a regression problem for a beer dataset (40 training samples) and a classification problem for a waste lubricant oil dataset (273 training samples). The results show that, after exhaustive pre-processing selection, iPLS variants show better performance for the first case study and remain competitive in the second case study. Wavelet transforms proved to be a viable alternative to classical pre-processing, improving performance for both linear and CNN models while maintaining interpretability. For the second case study, with more data, CNNs present good performance when applied on raw spectra and could potentially be used to avoid exhaustive pre-processing selection. However, it was found that CNNs can benefit from some types of pre-processing, leading to improved performance for the first case study and overall better performance for the second case study.

Significance and Novelty:

This study provides a critical and exhaustive comparison of combinations of pre-processing methods and models for spectroscopic data analysis. It was found that no single combination of pre-processing and model that can be identified as optimal beforehand in low data settings.

1. Introduction

Spectral data have been extensively used to predict the quality and quantity of substances in several fields, such as agricultural, pharmaceutical, food & beverages, and petrochemical [1,2]. Some recent reviews of chemometric methods are available in [3,4]. The multivariate calibration models used in spectroscopic analysis can be generally grouped into linear or non-linear methods. Among the class of linear models, Partial Least Squares (PLS) has been widely used to develop inferential models in industrial scenarios due to its ability to manage collinearity, noise, and a reasonable amount of missing data [3–6]. Additionally, the class of penalized regression methods, which includes the Least Absolute Shrinkage and Selection Operator (LASSO) [7], are also adapted to handle high-dimensional data [8]. Linear models can be however heavily affected by unwanted artifacts, such as light scattering effects, matrix background or electric noise. To overcome this limitation, spectral pre-processing techniques are usually employed

for noise reduction, base line correction and feature extraction [9,10]. The most common pre-processing techniques are Standard Normal Variate (SNV) [11], Multiplicative Scatter Correction (MSC) [12], and the Savitzky-Golay-Derivative (SGD) filter [13]. A further review of pre-processing techniques can be found in [9,10]. Due to the interconnection between pre-processing and model building, spectroscopic analysis usually requires an extensive search between different combinations of models and pre-processing methods [14]. Recent work also shows that there may be benefits in jointly optimizing the pre-processing parameters (e.g., SGD window size and derivative order) and model hyperparameters (e.g., number of PLS latent variables) [15].

An alternative, less used, pre-processing approach involves the application of wavelet transforms [16,17]. Wavelet transforms amount to filter banks that, by convolution with the original signals, decompose them into their time-frequency components (in the context of spectral

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data, the time domain can be replaced by the wavenumber or wavelength domains). The use of wavelet transforms in chemometrics can be traced back to the 1990s [18–23]. An introduction to multivariate calibration using wavelet transforms and PLS is given in [24], while signal compression and denoising is discussed in [25]. More recently, wavelet transforms have been used in several fields for decomposition of spectral data and identification of relevant features [26–29] and signal denoising by either: *i*) removal of the high frequency wavelet coefficients [30–32], *ii*) model-free or model-based (Kalman) filtering [33–36], *iii*) coefficient selection via soft-thresholding [37,38] or genetic algorithms [39,40]. As the wavelet coefficients retain the same information as the original signal, they can potentially be used as informative features from the high-dimensional spectra. Thus, in this work, wavelet transforms are first used as pre-processing tools for feature extraction followed by feature selection using forward and backward selection variants of interval PLS [41] and LASSO.

Recently, there has been a growing interest in the application of Deep Learning (DL) methods, most commonly, Convolutional Neural Networks (CNN), for modeling and analysis of spectroscopic data [31, 42–48]. CNN have several benefits, including the flexibility to model complex non-linear relationships, data fusion, transfer learning and can outperform traditional chemometric approaches in terms of accuracy, especially for larger datasets [48]. One key advantage of CNN is their ability to perform automatic pre-processing, or feature extraction, of the raw spectral data. This can reduce the cumbersome task of conducting an exhaustive search between multiple combinations of modeling and pre-processing methods. Recent works have also assessed the benefit of applying pre-processing before using CNNs [31,46]. In particular, it was found that traditional spectral pre-processing methods can increase the CNN accuracy for small datasets [46] and that wavelets can reduce noise in the spectral data and lead to clearer and more robust interpretability [31].

This study aims at assessing and comparing the efficacy of classical linear modeling approaches and the more recently proposed non-linear methodologies, through two distinct case studies. The first case study involves a regression analysis aimed at predicting a beer quality parameter using data derived from visible and near-infrared spectroscopy (VIS-NIR). The second case study addresses a binary classification problem, leveraging mid-infrared (MIR) spectroscopy to predict the coagulation potential of waste lubricant oils (WLO) during an industrial processing operation. Five classes of methods were evaluated: PLS with chemometric pre-processing; interval PLS with chemometric pre-processing or wavelet transforms; LASSO with wavelet transforms; and CNN applied to raw data, or combined with chemometric and wavelet transforms pre-processing. As an alternative to conventional pre-processing techniques, wavelet transforms were used to decompose the spectral data into multiple frequency components. Subsequently, feature selection, more specifically, the selection of wavelet coefficients, was performed using either LASSO or PLS-based modeling approaches. For the non-linear models, CNNs were applied to both raw and pre-processed spectra, including different types of chemometric and wavelet pre-processing approaches. This analysis enabled a thorough evaluation of the benefits of applying prior pre-processing when using CNNs, a commonly debated topic in this field [43,46,48]. Relevant information was also gathered on the potential add-value of using different types of modeling methodologies currently available.

The rest of this article is organized as follows. Section 2, presents the description of the datasets, pre-processing techniques, modeling methodologies, and the approach followed for model comparison and selection. Afterwards, in Section 3, we report the main results for the two real world case studies, as well as a general overview and discussion of the advantages and disadvantages of the various methods. Finally, Section 4 summarizes the main conclusions of this article.

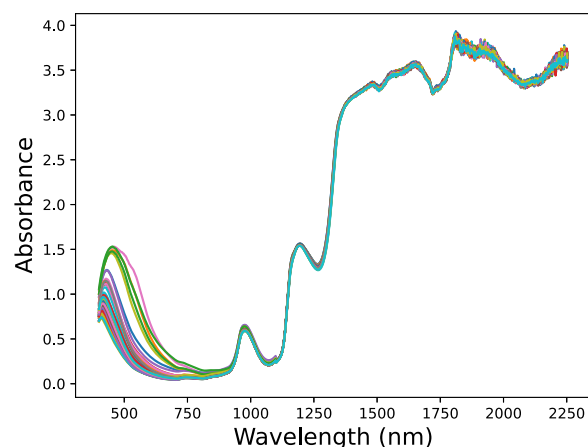


Fig. 1. VIS-NIR spectra related to beer case study.

2. Materials and methods

2.1. Datasets

2.1.1. Beer case study

The first case study regards a regression problem where a beer quality parameter, the real extract concentration, is predicted using VIS-NIR [5]. The dataset is composed by the real extract concentration (response) and the VIS-NIR spectra in the range of 400 nm to 2250 nm with a resolution of 2 nm, for a total of 926 predictors. The original dataset is divided into 40 training and 20 test samples, respectively. The spectra are presented in Fig. 1.

2.1.2. Waste lubricant oil case study

The second case study is related to the analysis of WLO, as part of the circular economy of lubricating oils managed by Sogilub, a Portuguese non-profit organization. During the regeneration process, WLO may coagulate due to different physico-chemical phenomena, which can lead to operational issues that should be avoided for safe and efficient operation. Further information can be found in [6]. To mitigate operational problems, the WLO samples are preliminary tested in the laboratory and classified into either a “positive” or “negative” class, leading to a binary classification problem.

A total of 109 WLO samples were analyzed using MIR spectroscopy within the spectral range of 4000 cm^{-1} to 500 cm^{-1} with a resolution of 2 cm^{-1} , resulting in 1814 predictors. From these WLO samples, 66 samples were from the negative class, and 43 samples were from the positive class. Each sample was analyzed in triplicate, leading to a total of 327 spectral replicates. The triplicates of each sample were treated as different observations, but they were always allocated to the same dataset. As the original dataset presents a class imbalance, a test dataset was constructed by randomly selecting 20% of the positive samples and an equal number of negative samples. Thus, the test dataset contains 54 spectra from 18 samples in triplicate. The training dataset contains 273 spectra from the remaining 91 samples. The raw spectra for this case study are presented in Fig. 2.

2.2. Spectral pre-processing

2.2.1. Chemometric approaches

In this study, we consider four widely used chemometric pre-processing techniques: Mean Centering (MC), SNV [11], MSC [12], SGD [13]. The simplest pre-processing technique is MC, where each individual spectra is subtracted by the mean of the training data.

SNV normalizes each sample using the mean and standard deviation computed over the wavelength or wavenumber dimension. More

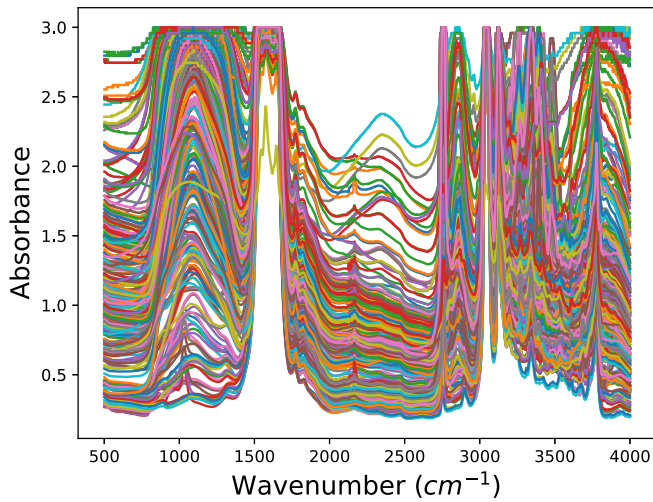


Fig. 2. Mid-infrared spectra of the samples from the WLO case study.

specifically, the corrected version of the i th spectrum is given by

$$\tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i - m_i}{s_i}, \quad (1)$$

where m_i is the mean of $\mathbf{x}_i$ and s_i is the standard deviation of $\mathbf{x}_i$.

MSC assumes that the spectrum of each sample, $\mathbf{x}_i$, can be properly modeled by:

$$\mathbf{x}_i = a_i + b_i \mathbf{x}_{ref} + \mathbf{e}_i, \quad (2)$$

where $\mathbf{x}_i$ is the raw spectrum for the i th sample, $\mathbf{x}_{ref}$ is a reference spectrum, and $\mathbf{e}_i$ is an unmodeled component. In this formulation, a_i represents the scatter additive effect and b_i represents the multiplicative effect. Both a_i and b_i are unknown parameters and may differ from sample to sample. The $\mathbf{x}_{ref}$ is typically set to the average spectrum of the training dataset. Afterwards, the estimates for a_i and b_i ($\hat{a}_i$ and $\hat{b}_i$) in each sample are determined by least squares regression. The corrected spectrum ($\tilde{\mathbf{x}}_i$) for the i th sample is then given by:

$$\tilde{\mathbf{x}}_i = \frac{\mathbf{x}_i - \hat{a}_i}{\hat{b}_i}. \quad (3)$$

SGD consists of fitting a low-order polynomial on a window of measurements centered at the wavelength or wavenumber of interest and, simultaneously, computing the derivative at the center point, which is saved for modeling. The same operation is applied to all data points of the raw spectrum. There are several hyperparameters in SGD: *i*) the order of the polynomial, *ii*) the size of the window over which the polynomial is fit, and *iii*) the order of the derivative to compute. In this work, a second degree polynomial is considered with three combinations of different window sizes and derivatives, as well as combination with SNV, as presented in Table 1.

2.2.2. Wavelet transforms

A family of wavelet basis functions can be generically represented as [49],

$$\psi_{s,u}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-u}{s}\right), \quad (4)$$

where t is the functional index, s and u are the dilation and translation parameters, and $\psi(t)$ is the mother wavelet. In the context of spectral data, the functional index t represents the wavenumber or wavelength domains. In the continuous wavelet transform (CWT) the scale and translation parameters vary continuously, constituting a redundant transformation. To apply the transform to discrete signals the scale and translation parameters are typically descritized dyadically as $s = 2^j$ and $u = k2^j$, for $(j, k) \in \mathbb{Z}^2$. Thus, for the discrete wavelet transform (DWT),

the wavelet function is rewritten as:

$$\psi_{s,u}(t) = \psi_{j,k}(t) = 2^{-j/2} \psi(2^{-j}t - k), \quad (5)$$

and it is coupled with a scaling function, obtained from a father scaling function $\phi(t)$, that captures the low frequency content of the signal:

$$\phi_{j,k}(t) = 2^{-j/2} \phi(2^{-j}t - k). \quad (6)$$

The family of wavelet functions is then generated by varying the dilation and translation parameters. Several wavelet filters can be found in the literature [16,49,50], being the Haar, Daubechies and Symlets wavelets the most common. As an example, for the Haar wavelet (the simplest wavelet function), the wavelet and scaling functions are defined as:

$$\psi(t) = \begin{cases} 1, & 0 \leq t < 1/2 \\ -1, & 1/2 \leq t < 1 \\ 0, & \text{otherwise,} \end{cases} \quad (7)$$

$$\phi(t) = \begin{cases} 1, & 0 \leq t < 1 \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

The DWT can be efficiently performed using the recursive algorithm proposed by [17], which is schematically represented in Fig. 3.

This procedure is initialized with $\mathbf{a}_0 = \mathbf{x}$, where $\mathbf{x}$ ($M \times 1$) is the original signal. Then, $\mathbf{a}_0$ is convoluted with a low pass filter ($\mathbf{H}$; related to the low frequencies) and a high pass filter ($\mathbf{G}$; related to the high frequencies). The outputs are then dyadically downsampled, producing a set of approximation ($\mathbf{a}_1$) and detail ($\mathbf{d}_1$) coefficients of the next decomposition level. The procedure is then repeated by further decomposing the approximation coefficients until a pre-established decomposition depth (J) is attained. Mathematically, the procedure is represented by recursive application of:

$$\begin{cases} \mathbf{a}_j = (\mathbf{H} * \mathbf{a}_{j-1}) \downarrow 2 \\ \mathbf{d}_j = (\mathbf{G} * \mathbf{a}_{j-1}) \downarrow 2 \end{cases} \quad (9)$$

where $*$ is the convolution operator and $\downarrow$ is the downsampling operator. For the case of the Haar wavelet $\mathbf{H} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$ and $\mathbf{G} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$.

Wavelet-thresholding techniques are effective for reducing the influence of noise in the signal. This can be achieved by setting the noisy, low-magnitude, wavelet coefficients (typically those related to high frequencies, *i.e.*, lower scales) to zero followed by signal reconstruction using the inverse transform operation.

In this study, the wavelet decomposition is applied in two distinct ways. For the case of iPLS and LASSO models, the DWT is used to decompose the spectrum into wavelet coefficients and implicitly extract features at different scales. Afterwards, the wavelet coefficients are fed to the modeling methodologies for variable selection. On the other hand, for the CNN models, the wavelet transforms are used for signal denoising by removing the wavelet coefficients corresponding to high-frequency noise. Then, the signal is reconstructed and fed to the CNN model. An illustration of this approach is presented in Fig. 4. The lowest scale d_1 , related to high-frequency components, is always eliminated. The same decomposition depth (J) is considered for both approaches. The wavelet transforms were implemented using the Wavelet Toolbox available in Matlab.

2.3. Modeling methodologies

2.3.1. Partial least squares

In general, for a regression problem, the PLS [51] derives a model between the spectrum ($\mathbf{X}$) and one or more responses ($\mathbf{Y}$) assuming a latent variable structure of the type:

$$\begin{cases} \mathbf{X} = \mathbf{TP}^T + \mathbf{E} \\ \mathbf{Y} = \mathbf{TQ}^T + \mathbf{F} \end{cases} \quad (10)$$

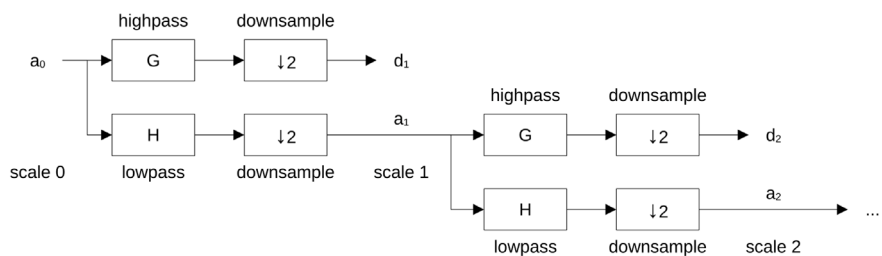


Fig. 3. Schematic representation of the recursive wavelet decomposition. $\downarrow 2$ represents a dyadic downsampling.

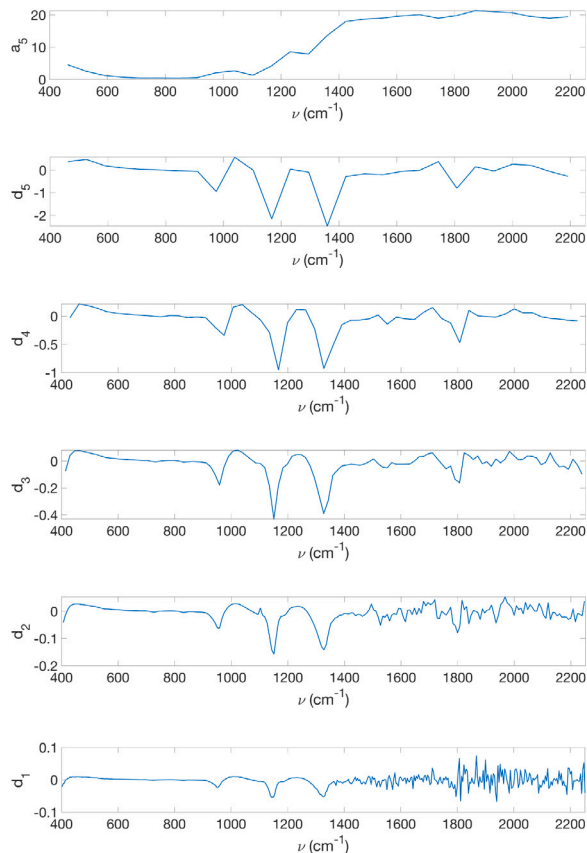


Fig. 4. Example of the wavelet decomposition of one spectrum from the beer dataset. In wavelet-thresholding, the spectrum is reconstructed after eliminating the wavelet coefficients below a specified scale and retaining all the others, for example, retaining the wavelet coefficients from scales d_3 up to d_5 and a_5 .

where, $\mathbf{X}$ is a $(N \times M)$ matrix of pre-processed spectra for N samples composed of M wavelengths or wavenumbers, $\mathbf{Y}$ is a $(N \times A)$ matrix of A response properties, $\mathbf{T}$ is a $(N \times K)$ matrix of latent variables, $\mathbf{P}$ ($M \times K$) and $\mathbf{Q}$ ($A \times K$) are loading matrices of $\mathbf{X}$ and $\mathbf{Y}$ -blocks, and $\mathbf{E}$ ($N \times M$) and $\mathbf{F}$ ($N \times A$) are residual matrices. The latent variables are estimated by $\mathbf{T} = \mathbf{X}\mathbf{W}(\mathbf{P}^T\mathbf{W})^{-1}$, where $\mathbf{W}$ is a $(M \times K)$ weighting matrix.

For classification problems, the $\mathbf{Y}$ response is modified to be an indicator column (with $A = 1$) that codify the class labels. In this case, PLS is applied to extract the lower-dimensional subspace with discriminative power, that is, to compress the spectra into a few discriminative latent variables ($\mathbf{T}$). These latent variables (or classification features) are then fed to the Bayesian linear classifier [12] for maximum discrimination. The Bayesian linear classifier provides an estimate of the

class label as the one that minimizes the expected classification cost:

$$\hat{y} = \operatorname{argmin}_l \sum_{t=1}^L \hat{Pr}(l|t)C(y|l) \quad (11)$$

where $\hat{y}$ is the predicted class, L is the number of classes, $\hat{Pr}(l|t)$ is the posterior probability of class l for observation t and $C(y|l)$ is the cost of classifying an observation as y when its true class is l . The posterior probability that an observation t is of class l is

$$\hat{Pr}(l|t) = \frac{Pr(t|l)Pr(l)}{Pr(t)}, \quad (12)$$

For this study, the prior $Pr(l)$ was set to the classes relative frequencies in $\mathbf{Y}$ and the cost is set to 0 for correct classifications and 1 for incorrect classifications. The number of latent variables, K , which acts as the hyperparameter of the PLS, is selected using Monte Carlo Cross Validation (MCCV) based on an hold-out set of data (see Appendix A). The PLS model was implemented using a custom Matlab code, using the NIPALS algorithm.

2.3.2. Interval partial least squares

By default, the standard PLS uses the entire spectrum to fit a model and thus it is prone to produce a sub-optimal model. To improve the performance of PLS, interval PLS (iPLS) [41] divides the spectra into several intervals and then builds a PLS model using only a subset of intervals with relevant information. The relevant intervals can be selected following either forward addition, through Forward iPLS (FiPLS), or backward elimination, through Backward iPLS (BiPLS), to minimize a loss function $\mathcal{L}$. For regression problems, $\mathcal{L} = RMSE$ and for classification problems $\mathcal{L} = -ACC$ (see Section 2.5 for the definition of $RMSE$ and ACC).

FiPLS starts by building a tentative PLS model for each interval using the MCCV procedure described in Appendix A. In this stage, for each tentative model, a loss function $\mathcal{L}$ is evaluated multiple times and stored in a $\mathbf{L}_{CV}$ vector. The interval with the lowest median $\mathbf{L}_{CV}$ is then selected as the initial reference model. In the next stage, the algorithm tentatively adds one interval at a time to the reference model and compares its $\mathbf{L}_{CV}$ with the reference. The comparison is made through a one-sided paired Wilcoxon signed rank test [52]. Among the intervals with statistically lower $\mathbf{L}_{CV}$, the interval leading to the lowest median $\mathbf{L}_{CV}$ is added to the reference model. This procedure is then repeated until adding intervals to the model no longer improves the $\mathbf{L}_{CV}$ in a statistically significant way.

In the case of BiPLS, the initial reference model contains all intervals. Afterwards, each interval is tentatively removed from the model using the same procedure as FiPLS. The algorithm stops when removing intervals from the model does not improve its performance.

In the case of iPLS using wavelet transforms, each spectrum is first decomposed into J scales using a DWT. The wavelet coefficients at each scale are then divided into non-overlapping intervals of dyadic length. With this division, a set of pseudo-intervals are obtained, where each pseudo-interval contains the wavelet coefficients of scale j in interval i . As before, the pseudo-intervals to retain in the model are selected using either FiPLS or BiPLS. The iPLS models were, similarly to the full PLS, implemented using custom Matlab code based on the NIPALS algorithm. In both implementations of FiPLS and BiPLS, the tolerance for the statistical test was set to 0.10 as proposed in [14].

2.3.3. Least absolute shrinkage and selection operator

LASSO [7] is a regularization method that removes redundant variables by reducing their regression coefficient to zero. This is done by adding a L_1 -norm penalty on the magnitude of the regression coefficients to the loss function. For the case of regression, the penalty term is added to the mean squared error, and thus the LASSO loss function becomes:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^M |\hat{b}_j| \quad (13)$$

where $\hat{\mathbf{b}} = [\hat{b}_1, \dots, \hat{b}_M]^T$ are the regression coefficients estimated by the LASSO model, and λ is a non-negative hyperparameter that controls model complexity. Likewise, for binary classification ($y_i \in \{0, 1\}$), LASSO fits a penalized logistic regression using the negative log likelihood criteria. Thus, the loss function for a classification problem becomes the Binary Cross Entropy (BCE):

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \ln(\hat{p}_i) + (1 - y_i) \ln(1 - \hat{p}_i)] + \lambda \sum_{j=1}^M |\hat{b}_j| \quad (14)$$

where $\hat{p}_i = (1 + e^{-z})^{-1}$, with $z = \hat{b}_0 + \mathbf{x}_i^T \hat{\mathbf{b}}$, is the predicted probability that the i th sample belongs to class 1.

When $\lambda = 0$ the penalty has no effect and LASSO resumes to the standard linear regression and logistic models. However, as λ becomes larger the regression coefficients are shrink towards zero. If λ is sufficiently large, then the null model where all regression coefficients are set to zero is obtained. Thus, the optimal λ is a balance between goodness of fit and model complexity. As described in Section 2.5, λ is determined using grid search and Cross Validation (CV) by search over a grid of λ values (see Table 2). The LASSO method was implemented using the Statistics and Machine Learning Toolbox available in Matlab.

2.3.4. Convolutional neural network

In this work, a 1D CNN is used to model the spectral data. A CNN consists of multiple layers which perform different mathematical operations, including convolutional, pooling, flattening and standard fully connected (FC) layers. The convolutional layer extracts features from data by, similarly to the wavelet methods described earlier, performing a convolution using a moving window filter throughout the input spectrum. However, in contrast to wavelets, which have specific fixed functional forms, the CNN relies on a linear weighted filter, or kernel, whose weights can be estimated from data. This provides the flexibility to automatically extract features tailored to each dataset and target variable. The filter uses a stride of 1 and SAME padding, where the output of the convolutional layer is padded with zeros so that its size is the same as the input signal. By considering multiple filters in parallel, different features can also be simultaneously identified in each convolutional layer. The extracted features are transformed using a non-linear activation function, such as Sigmoid, Hyperbolic Tangent (Tanh), Rectified Linear Unit (ReLU) or Exponential Linear Unit (ELU). A pooling layer performs dimensionality reduction and eliminates redundant features by retaining either the maximum or the average value in a sliding window of a given size across the signal. In this work, we treat the type of pooling as a categorical hyperparameter, allowing either maximum, average or no pooling at all, to be applied. Although the size of the pooling window is dependent on the input spectra dimension and could be treated as another hyperparameter, we choose to fix the size to avoid conditional hyperparameters. The pooling layer also uses a stride of 1. After the signal is vectorized using a flattening layer, FC layers and activation functions are recursively applied to relate the extracted features with the target variables for regression or classification problems. Several CNN structures for modeling of spectroscopic data can be found in the available literature, ranging from simpler [43] to more complex models inspired by CNN architectures used in computer vision tasks [44,46]. In this study, due to the relatively small datasets and to enable better interpretability, we

opt for a simpler architecture, consisting of only one convolution layer and filter, followed by either a maximum or average pooling layer or no pooling at all, as described earlier, in addition to multiple FC layers. Fig. 5 illustrates the CNN architecture used in this work.

CNN models can be more prone to overfitting than other simpler linear models, due to their comparatively much larger number of parameters. To mitigate this issue, several strategies can be used, including early stopping, dropout layers and regularization. In this work we apply the latter approach, where the loss function, $\mathcal{L}$, is penalized with the squared L_2 norm of all the model parameters (weights and biases), $\mathbf{w}$:

$$\mathcal{L}_\lambda = \mathcal{L} + \lambda \|\mathbf{w}\|_2^2, \quad (15)$$

where the regularization weight, λ , is a positive coefficient that penalizes model parameters with large values. Similar to the LASSO method described earlier, the loss function, $\mathcal{L}$ is the (unregularized) MSE or the BCE for regression or binary classification tasks, respectively. We note that other forms of regularization can also be used, including L_1 penalization similar to the LASSO method. In this work, the Adam optimization algorithm is used to optimize the model parameters, changing only the learning rate and keeping other optimizer parameters at their default values. Although mini-batch training is generally recommended in DL model development, we opt to use full batch training due to smaller dataset sizes. A total of 500 epochs is considered for the model training. Since the optimization landscape for the loss function in CNN is usually non-convex and multi-modal, the initialization of the model parameters can greatly affect the optimization solution and thus the model predictive performance. In this work, we initialize all layer weights with uniform Glorot initialization [53] when using Sigmoid and Tanh activation functions, and normal He initialization [54] for ReLU and ELU activation functions. The biases are all initialized as zero. The CNN models were implemented in Python, using PyTorch, and trained on a dedicated workstation with GPU acceleration.

The CNN has several hyperparameters, including those related to the optimizer (learning rate and regularization weight) as well as the network structure (type of pooling layer, kernel size, number and size of hidden fully connected layers and type of activation function). In addition, when using wavelet pre-processing, we also include the level of decomposition as an additional hyperparameter. This is done through thresholding, as described in Section 2.2.2. Since the lowest scale is always removed, we have an integer valued hyperparameter ranging from 2 up to the maximum scale number, J . In this work, these hyperparameters are selected based on CV, using Bayesian Optimization (BO) [55], which is a widely used approach for DL hyperparameter optimization [56,57]. Appendix B presents the algorithmic details of the proposed BO approach, which was implemented using the Python package Ax with Botorch backend [58]. Section 2.5 presents additional details on the hyperparameters and their ranges (see Table 2) considered in this work.

2.4. Model interpretability

An important factor to consider in the analysis and interpretation of models for spectral data is the importance of each feature, in this case the measurements across different wavelengths or wavenumbers, which can be linked to the presence of specific chemical compounds.

In the previously mentioned linear modeling approaches, feature importance can be easily quantified by looking at the values of the model coefficients. Additionally, for PLS models with a single response variable, the Variable Importance in Projection (VIP) [59] for the feature x_i can also be calculated as [60]:

$$VIP(x_i) = \sqrt{\frac{M \cdot \sum_{k=1}^K (q_k^2 \mathbf{t}_k^T \mathbf{t}_k) \cdot (w_{i,k} / \|\mathbf{w}_k\|)^2}{\sum_{k=1}^K (q_k^2 \mathbf{t}_k^T \mathbf{t}_k)}} \quad (16)$$

where $\mathbf{t}_k$ is the k th column of the matrix of latent variables $\mathbf{T}$, q_k is the k th element of the loadings matrix $\mathbf{Q}$, $\mathbf{w}_k$ is the k th column

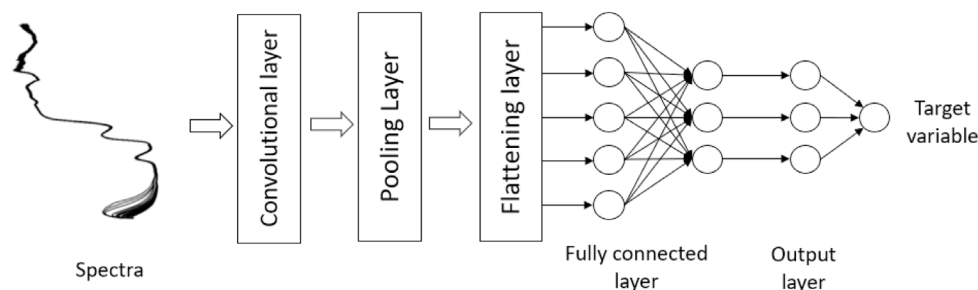


Fig. 5. CNN model architecture considering only a single hidden fully connected layer.

of the weighting matrix $\mathbf{W}$ and $w_{i,k}$ is the i th element of vector $\mathbf{w}_k$. The $w_{i,k}/\|\mathbf{w}_k\|$ represents the importance of the i th variable in the k th latent variable. The $q_k^2 \mathbf{t}_k^T \mathbf{t}_k$ represents the variance of $\mathbf{Y}$ explained by the k th latent variable and the summation of these terms over all K components is the total variance explained by the PLS model. The sum of squared VIP is equal to the number of predictor variables (i.e., $\sum_{j=1}^M VIP_j^2 = M$). Typically, variables with VIP larger than 1 are considered the most relevant to explain the response variable [60].

Contrary to simpler linear modeling approaches, the relationship between input and output variables in DL models is much more complex and can generally be seen as a black-box. In an effort for eXplainable Artificial Intelligence (xAI), different methods have been recently proposed to increase the interpretability of black-box models, including DL [61]. Several methods can be used to determine feature importance by considering either DL specific gradient-based approaches such as integrated gradients (IG) [62] or Grad-CAM [63] or model agnostic methods such as SHAP [64] or LIME [65]. In this work, we use IG, a gradient based method which can be applied for both regression and binary classification cases. Using this method, the importance $IG(x_i)$, of a given feature, x_i , is given by:

$$IG(x_i) = (x_i - x'_i) \int_0^1 \frac{\partial F(x'_i + \alpha(x_i - x'_i))}{\partial x_i} d\alpha \quad (17)$$

where $IG(x_i)$ is the IG value, where higher values reflect greater importance of feature x_i , based on the deviation from a baseline input vector, x'_i , typically set to zero to represent the absence of a feature. $F(x)$ is the CNN predicted value for the target variable whose gradients, despite having a complex algebraic form, can be obtained using automatic differentiation methods available in standard DL software packages. The IG method is implemented using the Python package Captum. The IG values are calculated based on the input spectra, thus the feature importance is obtained considering the pre-processing performed by the convolutional layer. Unlike the VIP in PLS or LASSO model coefficients, the IG is calculated for each individual spectrum. It is worth to note that due to the stochastic nature of the CNN weight initialization, both the extracted features and their estimated importance may vary for two models trained under identical sets of hyperparameters. Thus, because we consider random weight initialization, either from a uniform Glorot or Normal He distribution, we train the model with the same set of hyperparameters 30 times, repeat the feature importance attribution for each trained model and average the results for each individual spectrum.

2.5. Model comparison and selection

In order to perform a thorough and exhaustive comparison between different approaches, a total of 51 combinations of modeling methodologies and pre-processing approaches were considered for both case studies. These include combinations of: nine standard chemometric techniques and three PLS methods (total of 27 models); five wavelet transforms and three feature selection methods, FiPLS, BiPLS and LASSO (total of 15 models) and CNN models applied both on raw spectra and with pre-processing (total of 9 models). For this last

Table 1

Overview of the different pre-processing techniques and modeling methods used in this work.

Pre-processing	Model			
	PLS	iPLS	LASSO	CNN
No pre-processing				✓
MC	✓	✓		✓
SNV	✓	✓		✓
MSC	✓	✓		✓
SGD-{1,7} ^a	✓	✓		
SGD-{1,15} ^a	✓	✓		
SGD-{2,9} ^a	✓	✓		✓
SNV-SGD-{1,7} ^a	✓	✓		
SNV-SGD-{1,15} ^a	✓	✓		
SNV-SGD-{2,9} ^a	✓	✓		✓
Haar-C ^b		✓	✓	
Dd4-C ^b		✓	✓	
Dd6-C ^b		✓	✓	
Sym4-C ^b		✓	✓	
Sym6-C ^b		✓	✓	
Haar-R ^b				✓
Dd6-R ^b				✓
Sym6-R ^b				✓

^a The short name of the Savitzky-Golay differentiation pre-processing is coded as SGD-{D,W}, where D is the derivative order and W is the window size.

^b Wavelet transforms can be applied in two distinct forms: as a reconstructed spectrum, denoted as “R”, or as wavelet coefficients, denoted as “C”.

approach, since the CNN can also perform some feature extraction and pre-processing through the convolutional filter, we consider a less extensive set of pre-processing methods, with five chemometric pre-processings and three wavelet transforms, for a total of 9 methods. An overview of the pre-processing approaches and modeling methodologies considered in this work is presented in Table 1.

The relative performance of all models under comparison in this work was evaluated using the autoML approach called SS-DAC (Soft Sensor Development Assessment and Comparison) [14]. SS-DAC is composed by three stages regarding (i) model building, (ii) model assessment and (iii) model comparison. The three stages are illustrated in Fig. 6.

In Stage I (model building), the raw dataset is split into training and test datasets. In this stage, the training dataset is used to select the model hyperparameters of the different models. For the case of PLS-based models, the hyperparameters were selected by MCCV using 400 random partitions of the training dataset into calibration and validation datasets. In each randomization, the calibration dataset contains 80% of the training samples selected at random, while the corresponding validation dataset contains the remaining samples. On the other hand, the hyperparameters for the CNN models were selected using 10-fold CV, by either minimizing the Root Mean Squared Error (RMSE) or maximizing the accuracy in the case of regression or classification, respectively. This optimization is performed using BO, as described in Appendix B. For the LASSO models, the λ hyperparameter was selected by grid search over a range of λ values and minimization of either the

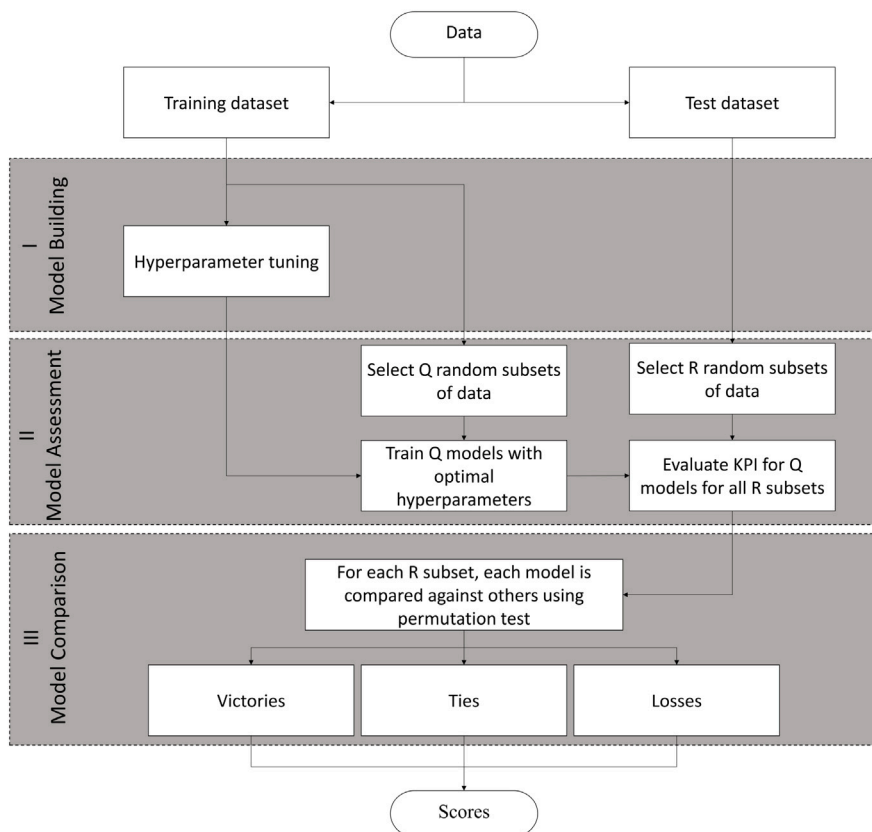


Fig. 6. Diagram of the SS-DAC stages for model building, assessment, and comparison.

Table 2
Hyperparameters of the different modeling methods. * indicates fixed values.

Model	Hyperparameters	Type	Search space (Beer)	Search space (WLO)
CNN	Learning rate	Continuous	$[10^{-4}, 10^{-2}]$	$[10^{-4}, 10^{-2}]$
	Regularization weight	Continuous	$[10^{-5}, 10^{-1}]$	$[10^{-5}, 10^{-1}]$
	Kernel size	Integer	[32, 128]	[48, 256]
	Hidden layers	Integer	1*	[1, 3]
	Hidden units	Integer	[10, 100]	[10, 100]
	Activation function	Categorical	[ReLU, ELU, Sig., Tanh]	[ReLU, ELU, Sig., Tanh]
	Pooling type	Categorical	[No pool, Max., Avg.]	[No pool, Max., Avg.]
	Pooling size	Integer	3*	5*
	Epochs	Integer	500*	500*
	Convolutional layers	Integer	1*	1*
LASSO	Regularization weight	Continuous (grid search)	$[10^{-5}, 0.5]$	$[10^{-5}, 1]$
PLS	Latent variables	Integer	[1, 926]	[1, 1814]
iPLS	Number of latent variables	Integer	[1, 926]	[1, 1814]
	Number of intervals	Integer	30*	30*
	Measurements per interval	Integer	31*	117*
	Number of intervals (wavelets)	Integer	29*	29*
	Decomposition depth (wavelets)	Integer	5*	5*
	Dyadic length (wavelets)	Integer	32*	64*

MSE or the BCE depending on the type of problem. For the classification case study, the data partitions used by all models were stratified to avoid imbalanced datasets.

In Stage II (model assessment) of SS-DAC, the models' performance were assessed over $Q = 200$ random training subsets and $R = 20$ random test subsets. The random training subsets allow for the estimation of performance variability related to fluctuations of the models' parameters. In turn, by using random test subsets it is also possible to estimate the expected performance variability when the models are applied to new data, which may be underestimated if a single test dataset was used. Both Q training and R test subsets are generated using bootstrapping of the original training and test datasets. To estimate both

sources of performance variability, each model is first retained in each $q = 1, \dots, Q$ training subset and their performance is then evaluated in each $r = 1, \dots, R$ test subset using a suitable Key Performance Indicator (KPI). From this stage, a total of $Q \times R$ realizations of the KPI are obtained.

As this work addresses both regression and classification problems, the models' KPI is dependent of the problem at hand. For the case of regression problems, the KPI is taken as the root mean squared error (RMSE), defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}, \quad (18)$$

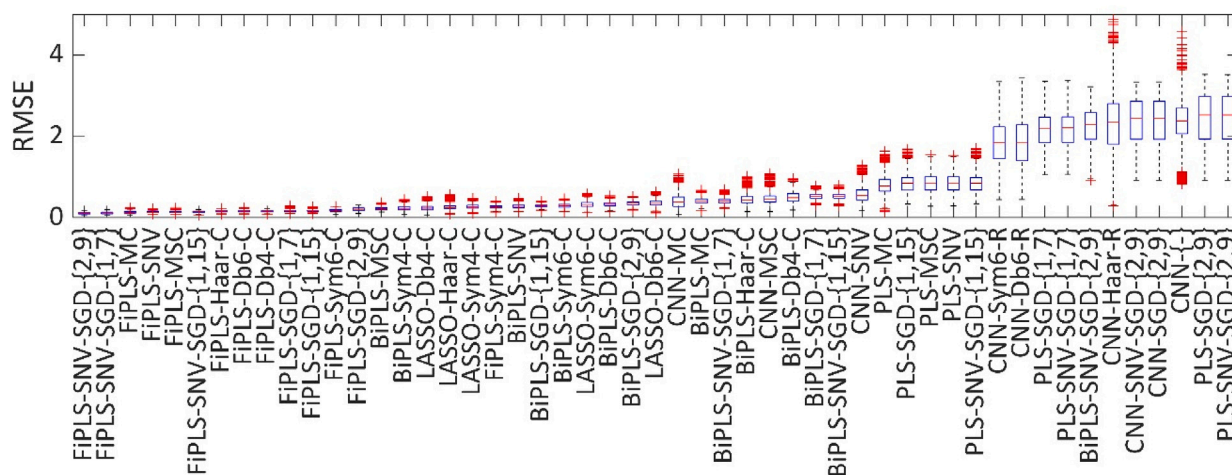


Fig. 7. RMSE of the models ranked from highest to lowest score using SS-DAC for the beer case study.

where y_i is the measured value, $\hat{y}_i$ is the model prediction for the i th sample and N is the number of samples. Likewise, for classification problems, the KPI is the accuracy (ACC), which is determined as,

$$ACC = \frac{TP + TN}{TP + FP + TN + FN} \quad (19)$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

In Stage III (model comparison) of SS-DAC the models' performance was compared using a permutation test [66] for each pair of models. These statistical tests are segregated by test subsets. That is, each model is compared against $(m - 1)$ other models in each R test subset, where m is the number of models. Based on the outcome of this statistical test, a *victory*, *tie* or *loss* is assigned to each model using the following definitions:

1. Victory: if the model has statistically better KPIs;
2. Tie: if there is no statistically significant difference in the models' KPIs;
3. Loss: if the model has statistically worse KPIs.

Afterwards, a score is attributed to each model by summing up the number of victories and ties, and the model's rank is obtained by sorting their scores from higher to the lowest.

3. Results and discussion

The code for the first case study, along with specific software versions, is available in https://github.com/PCLabUC/model_preprocessing_comparison_spectral_analysis. Although the data and code for the WLO case study are proprietary, both case studies were analyzed using the same software versions.

3.1. Beer case study

Fig. 7 presents the distributions of all RMSE for $Q \times R$ randomizations of the dataset obtained in Stage II of SS-DAC ranked by the performance score obtained in Stage III of the SS-DAC. In Table 3 the best model within each class and the CNN applied on raw spectra are presented, along with the selected hyperparameters. The hyperparameters of the remaining models are provided in Appendix C.

For this case study, all FiPLS models outrank the other models, both when combined with standard chemometric pre-processing and wavelet transforms. In particular, the combination of FiPLS and SNV-SGD results in the best overall model. These models are followed by three of the five LASSO variants. This indicates that the proposed wavelet

approach presents a good performance for this case study. CNN applied on the raw spectra leads to bad performance, being one of the lowest ranked models and with large variability. Applying pre-processing improves performance, with all CNN pre-processing combinations being better ranked. However, for some CNN models, mainly those using wavelet transforms and SNV-SGD, the improvement is not significant. For this case study, the simpler MC pre-processing leads to the best CNN model, which presents better performance than all full spectra PLS models. In particular, this latter class of models are consistently in the lowest rankings, with similar variability as the CNN on raw data training with stochastic weight initialization.

Fig. 8 presents the feature importance, according to the methods described in Section 2.4. for each class of models discussed. The top FiPLS models using both chemometric and wavelet pre-processing correctly eliminate the wavelengths above 1500 nm, which are considered noisy and irrelevant, and the wavelengths below 900 nm which are systematic but irrelevant [5]. Likewise, even though the LASSO models can do a point selection of wavelet coefficients, when the wavelet filter is taken into account, it is verified that higher wavelengths are still mostly represented in the model. In spite of these shortcomings, the models were able to identify the wavelengths between 1100 nm and 1350 nm as the most relevant. For the best ranked CNN model, CNN-MC, the feature importance analysis reveals that despite the disregard of the noisy irrelevant band, this model presents more variability in the identification of the relevant 1100–1350 nm band. Interestingly, this can lead to the conclusion that, despite improving predictive performance, the use of pre-processing can also limit, to an extent, the capability of the CNN to capture the relevant features of the spectra. Because the L_2 regularization is being used for the CNN, no explicit feature elimination is performed, which leads to a retention of all bands of the spectra, unlike LASSO or iPLS model. Contrary to the other models, the standard PLS leads to poor results in terms of interpretability, with the systematic but irrelevant bands below 700 nm being identified as important. Overall, the results are well aligned with those presented in the original work of [5], which found that methods incorporating variable selection perform better in this dataset.

3.2. Waste lubricant oil case study

Fig. 9 presents the distributions of all ACC for the $Q \times R$ randomizations obtained in the Stage II of SS-DAC ranked by the performance score obtained in Stage III of SS-DAC. In Table 4 the best model within each class with highest performance are presented, along with the selected hyperparameters. The hyperparameters of the remaining models are provided in Appendix D.

Table 3

Results for the best model of each class for the beer case study. The values presented are the median and interquartile range, in brackets, for all the RMSE obtained in SS-DAC Stage II.

Ranking	RMSE	Model	Hyperparameters	
1	0.11 (0.03)	FiPLS-SNV-SGD-{2,9}	Number of LVs	10
			Number of selected features	62
7	0.16 (0.03)	FiPLS-Haar-C	Number of LVs	12
			Number of selected features	25
16	0.24 (0.09)	LASSO-Db4-C	Regularization weight	4.32×10^{-4}
27	0.40 (0.23)	CNN-MC	Learning rate	0.0095
			Regularization weight	1×10^{-5}
			Kernel size	128
			Hidden units	90
			Activation	ELU
			Pooling type	Max.
36	0.79 (0.28)	PLS-MC	Number of LVs	12
			Number of selected features	25
49	2.39 (0.63)	CNN-raw	Learning rate	0.0012
			Regularization weight	0.0868
			Kernel size	128
			Hidden units	45
			Activation	ReLU
			Pooling type	No

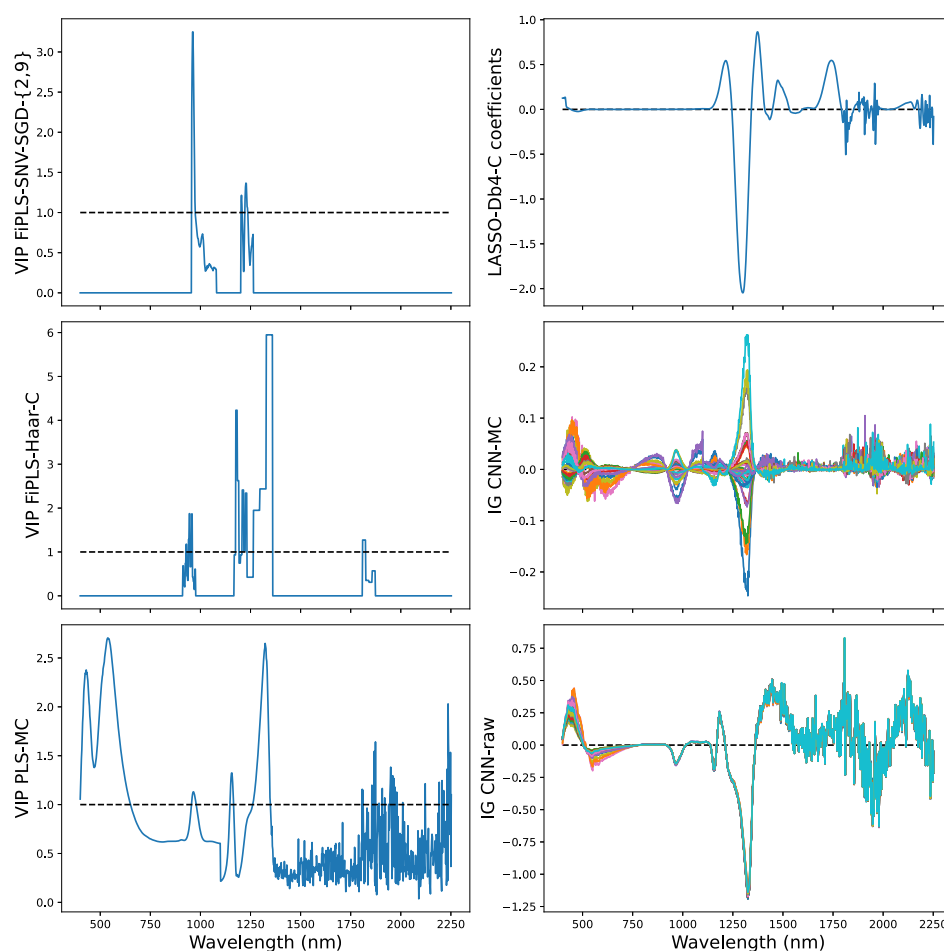


Fig. 8. Feature importance for the best models in each class for the beer case study. For the CNN models, the feature importance is presented for each spectrum in the training dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

A high median accuracy, in the range of 0.88 to 1, was attained by the best model within each modeling classes. CNNs leads the ranking of the different models, with a total of six variants, including CNN applied on raw spectra, on the top 10 models. In particular, the best overall model is a CNN using the Haar wavelet for denoising, with

a median ACC of 1, that is, a 100% correct label classification for 50% of the $Q \times R$ randomizations. Looking at the hyperparameters, it can be seen that using the Haar wavelet pre-processing with only the first scale being eliminated led to BO identifying a smaller, less parameterized, CNN model, with higher regularization weight, leading

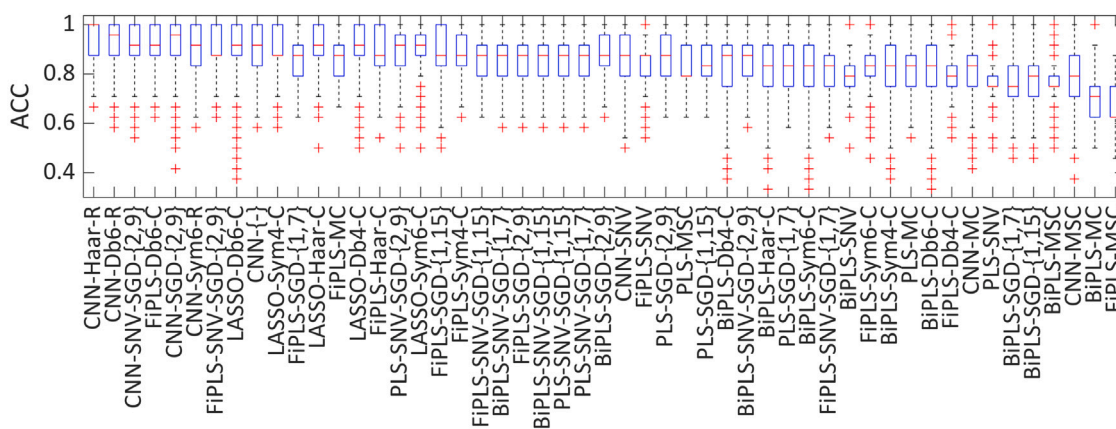


Fig. 9. Accuracy of the models ranked from highest to lowest score using SS-DAC for the WLO case study.

Table 4

Results for the best model of each class for the WLO dataset. The values presented for the accuracy are the median and interquartile range (in brackets) in the test dataset.

Ranking	Accuracy	Model	Hyperparameters	
1	1.00 (0.13)	CNN-Haar-R	Learning rate	1.2×10^{-3}
			Regularization weight	3.3×10^{-2}
			Kernel size	48
			Hidden layers	3
			Hidden units	10
			Activation	Tanh
			Pooling type	No
			Lowest wav. scale ret.	2
4	0.92 (0.13)	FiPLS-Db6-C	Number of LVs	11
			Number of selected features	358
7	0.88 (0.13)	FiPLS-SNV-SGD-{2,9}	Number of LVs	6
			Number of selected features	363
8	0.92 (0.13)	LASSO-Db6-C	Regularization weight	1.63×10^{-4}
9	0.92 (0.17)	CNN-raw	Learning rate	0.0008
			Regularization weight	1×10^{-5}
			Kernel size	184
			Hidden layers	3
			Hidden units	82
			Activation	Tanh
			Pooling type	Max.
16	0.92 (0.13)	PLS-SNV-SGD-{2,9}	Number of LVs	15
			Number of selected features	1814

to a better generalization ability. This result indicates that, in this case, even simple Haar wavelet denoising can be used to improve the hyperparameter optimization for CNN models. When applied on raw spectra, CNN also shows competitive performance. Although wavelets and SNV-SGD improve the CNN performance, other pre-processing approaches lead to worse performance than not using pre-processing at all, particularly the MSC. As for the other models, FiPLS, both with chemometric pre-processing and wavelet transforms, LASSO and PLS present similar performance. Contrary to the other case study, the full spectra PLS presents better performance and is ranked higher.

Fig. 10 displays the feature importance for the different models. It can be seen that for all models, the 2600–1800 cm^{-1} band is excluded. The best PLS and iPLS models identify the band around 1775–1700 cm^{-1} as important, specially FiPLS-SNV-SGD-{2,9} which had almost no other band included in the model. FiPLS-Db6-C made a broader selection in the same region, indicating the 1790–1574 cm^{-1} band as relevant. Contrary to the results of the previous case study, the full PLS-SNV-SGD-{2,9} model presents a clearer interpretation of the important features. Within the bands consistently selected by these models, the 1750–1735 cm^{-1} band is associated to esters [67]. The magnitude of the LASSO coefficients is also high for the aforementioned band, although the results are not as clear.

On the other hand, the feature importance analysis for the CNN leads to a different interpretation. For both the CNNs using raw spectra and Haar wavelets, the band around 2700 cm^{-1} is clearly identified as the most important, which is related to $C = O$ stretch in hydrocarbons and is also present in a lower degree in esters [67]. This can indicate that there is a non-linear relation between the spectra intensity at this band and WLO coagulation behavior that is not captured by the linear models. However, because the CNN does not perform explicit feature selection, the IG interpretation is not as explicit as the FiPLS case. Even though the conclusions are similar for both CNN variants, the use of Haar wavelet pre-processing appears to reduce some high frequency components of the spectra and enables a clearer interpretation of the important features to the model predictions. Similar conclusions were also found in recent work combining wavelets and CNN [31]. Based on these results, it is conjectured that esters may have a significant effect on the WLO coagulation phenomenon.

3.3. Discussion

Based on the results of the case studies considered in this work, several conclusions can be identified. Although these cannot be generalized for every possible case study, the authors identify some

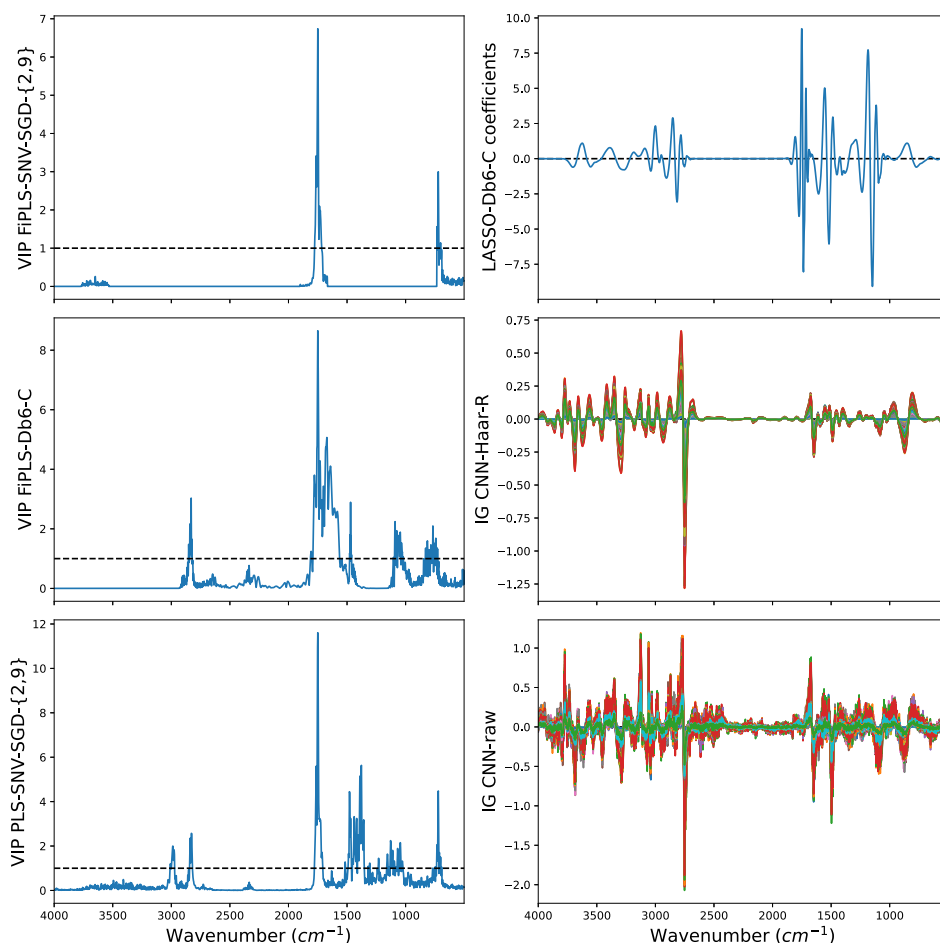


Fig. 10. Feature importance for the best models in each class for the WLO case study. For the CNN models, the feature importance is presented for each spectrum in the training dataset. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

guidelines that could potentially be used for other types of problems, more specifically those with a smaller dataset size.

Interval-based PLS models using adequate pre-processing present competitive performance.

The results indicate that interval PLS models, particularly those employing the forward selection algorithm (*i.e.*, FiPLS), have competitive performance on the WLO dataset and surpass the CNN model in the beer dataset. This superior performance can be attributed to the ability of FiPLS to effectively exclude irrelevant features, thereby enhancing model generalization and interpretability.

Wavelet transforms are a solid alternative pre-processing methodology for spectroscopic analysis.

Our findings suggest that wavelet transforms represent a robust alternative to traditional pre-processing methods, owing to their capability to effectively filter signals. The integration of FiPLS or LASSO with wavelet pre-processing reveals that, for the first dataset, characterized by reduced noise levels, there is a slight decrease in performance. However, for the second dataset, which exhibits higher noise levels, wavelet pre-processing results in a modest improvement in model performance. When combined with CNNs, wavelets consistently improved the predictive performance for both case studies.

CNN can lead to good performance on raw spectra but can also benefit from pre-processing.

In the WLO case study, with a few hundred samples in the training dataset, CNN on raw spectra led to a good performance, better than the

majority of the other models. This highlights the capability of automatic pre-processing and feature extraction that is usually presented as an advantage of CNN. When using xAI methods for feature importance analysis, CNNs, despite being viewed as black-box, can capture some of the relevant patterns even when applied on unprocessed spectra. However, this study also found that, in line with recent works [31,46], CNN can also benefit from pre-processing of the spectra. However, this is not always true for every pre-processing method, as for the WLO case study the use of some pre-processing methods led to worse performance.

No single pre-processing technique is optimal; the optimal pre-processing technique depends on the data and is interconnected with model selection.

In this work no single pre-processing method stood out as a clear winner for all models. The optimal pre-processing is always case dependent and multiple alternatives should be considered during model development. Thus, pre-processing should ideally be treated as an integral part of model development, both for linear and DL methods. As recent works indicate [15], it can be beneficial to jointly optimize pre-processing parameters and model hyperparameters during model development and optimization. Additionally, the SS-DAC framework proved to be a viable tool to compare the performance of multiple models and aid in the identification of the best combination of pre-processing and modeling methodology.

Table C.1
Hyperparameters of the PLS-based models for the beer case study.

Class	Pre-processing	# Latent variable			# Selected features		
		PLS	FiPLS	BiPLS	PLS	FiPLS	BiPLS
Chemometric	MC	5	6	8	926	93	525
	SNV	4	6	9	926	123	401
	MSC	4	6	6	926	123	402
	SGD-{1,7}	6	10	6	926	92	433
	SGD-{1,15}	5	8	9	926	62	341
	SGD-{2,9}	1	10	4	926	62	402
	SNV-SGD-{1,7}	6	5	10	926	62	402
	SNV-SGD-{1,15}	5	8	9	926	93	310
	SNV-SGD-{2,9}	1	4	3	926	93	617
Wavelet	Haar-C	–	12	19	–	25	266
	Db4-C	–	14	13	–	16	178
	Db6-C	–	11	12	–	16	135
	Sym4-C	–	26	9	–	26	202
	Sym6-C	–	24	6	–	24	202

Table C.2
Hyperparameters of the LASSO models for the beer case study.

Pre-processing	$\lambda \times 10^4$	# of selected features
Haar-C	3.40	24
Db4-C	4.32	39
Db6-C	3.33	39
Sym4-C	3.40	21
Sym6-C	8.59	34

4. Conclusions

This work performs a critical analysis of the combination of several pre-processing methods with classic modeling approaches such as interval and full spectra PLS as well as LASSO, and modern deep learning methods such as CNNs for modeling of spectroscopic data. This analysis was performed in a low data regime, up to a few hundred samples, for both regression and classification problems. A feature importance analysis was also considered to assess model interpretability.

Based on the results of this work, several conclusions can be outlined. The use of wavelet transforms to pre-process the spectrum proved to be competitive against traditional pre-processing methodologies, and can reduce the search space during pre-processing selection. From the different modeling approaches, interval based PLS methods are a competitive approach for spectroscopic analysis. For the classification case study, with a few hundred samples, CNN without any pre-processing presented good performance, highlighting their ability to perform automatic pre-processing and feature extraction. Moreover, unlike some previously made claims, this study also found that, similar to classic methods, CNN can also benefit from applying pre-processing techniques, chemometric or wavelets, to the spectroscopic data before model building. However, not all pre-processing techniques improved CNN performance. Overall, this indicates that there is not a single optimal pre-processing technique that can be identified beforehand.

For future work, we would like to point the need to improve the selection criterion used by PLS models, as the current approaches tends to select a high number of latent variables. Furthermore, to reduce the burden associated with an exhaustive search over multiple pre-processing techniques and models, a more efficient approach, integrating pre-processing with model hyperparameter optimization, can be considered for PLS, LASSO and CNN models. Due to the large number of model and pre-processing combinations, Bayesian Optimization can be a potential solution to efficiently tackle this problem.

CRedit authorship contribution statement

Rúben Gariso: Writing – original draft, Methodology, Conceptualization. **João P.L. Coutinho:** Writing – original draft, Methodology, Investigation. **Tiago J. Rato:** Writing – original draft, Validation, Supervision. **Marco S. Reis:** Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ruben Gariso, João Coutinho, Tiago Rato and Marco Reis reports financial support was provided by SOGILUB - Sociedade de Gestão Integrada de Óleos Lubrificantes Usados, Lda. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the financial support of SOGILUB – Sociedade de Gestão Integrada de Óleos Lubrificantes Usados, Portugal, Lda. The authors also acknowledge the support from the Chemical Engineering and Renewable Resources for Sustainability (CERES), Portugal, which is financed by national funds from FCT/MCTES (reference UIDB/00102/2020). The authors also acknowledge support from the Fundação para a Ciência e Tecnologia, Portugal, I.P., through the projects with DOI:10.54499/UIDB/00102/2020 and DOI:10.54499/UIDP/00102/2020. Finally, the authors would like to thank the anonymous reviewers for the suggestions made to improve the manuscript.

Appendix A. Hyperparameter tuning in PLS

The hyperparameters of the PLS-based models were determined by Monte-Carlo Cross-Validation (MCCV) using the following iterative algorithm.

The algorithm starts by randomly split the training dataset into C pairs of calibration and validation datasets. For each randomization, the calibration dataset contains 80% of the training samples, while the validation dataset is composed of the remaining 20% of the training samples. In the first iteration, the number of latent variables is set to $k = 1$. Then, for each randomization ($c = 1, 2, \dots, C$), a PLS model is trained on the calibration dataset using k latent variables and its performance is evaluated in the matching validation dataset. The model performance is evaluated through a loss function $\mathcal{L}$ and it is saved in a $(C \times 1)$ vector: $\mathbf{L}_{CV}(k)$. The procedure is then repeated for an additional latent variable, returning $\mathbf{L}_{CV}(k + 1)$. For a regression problem $\mathcal{L} = RMSE$ and for a classification problem $\mathcal{L} = -ACC$.

The performances of the models with k and $k + 1$ latent variables are then compared using a one-sided paired Wilcoxon signed rank test:

$$\begin{aligned} H_0 &: \text{median}(\mathbf{L}_{CV}(k)) \leq \text{median}(\mathbf{L}_{CV}(k + 1)) \\ H_1 &: \text{median}(\mathbf{L}_{CV}(k)) > \text{median}(\mathbf{L}_{CV}(k + 1)) \end{aligned} \quad (\text{A.1})$$

Table C.3
Hyperparameters of the CNN models for the beer case study.

Hyperparameter	Raw	MC	MSC	SNV	SGD-2-9	SNV-SGD-2-9	Haar-R	Db6-R	Sym6-R
Lowest wav. scale retained	–	–	–	–	–	–	3	2	3
Hidden units	45	90	56	61	92	89	78	74	100
Kernel size	128	128	33	128	38	126	35	101	50
Regularization weight	0.0868	1×10^{-5}	0.0010	0.0560	0.0120	0.0311	0.0597	0.0406	0.0282
Learning rate	0.0012	0.0095	0.0072	0.0052	0.0045	0.0060	0.0069	0.0073	0.0075
Pooling type	No	Max.	Avg.	Avg.	Max.	Max.	Avg.	Avg.	Max.
Activation function	ReLU	ELU	ELU	ELU	Sig.	Sig.	ReLU	Tanh	Tanh

Table D.1
Hyperparameters of the PLS-based models for the WLO case study.

Class	Pre-processing	# Latent variable			# Selected features		
		PLS	FiPLS	BiPLS	PLS	FiPLS	BiPLS
Chemometric	MC	9	8	2	1814	242	1572
	SNV	4	7	4	1814	363	1330
	MSC	4	4	3	1814	121	1693
	SGD-{1,7}	5	10	9	1814	121	1572
	SGD-{1,15}	7	10	7	1814	242	1330
	SGD-{2,9}	11	5	9	1814	363	1694
	SNV-SGD-{1,7}	5	10	5	1814	363	1693
	SNV-SGD-{1,15}	5	5	5	1814	484	1693
	SNV-SGD-{2,9}	15	6	5	1814	363	1693
Wavelet	Haar-C	–	12	19	–	538	1742
	Db4-C	–	14	13	–	224	1819
	Db6-C	–	11	12	–	358	1852
	Sym4-C	–	26	9	–	160	1789
	Sym6-C	–	24	6	–	274	1858

Table D.2
Hyperparameters of the LASSO models for the WLO case study.

Pre-processing	$\lambda \times 10^4$	# of selected features
Haar-C	2.31	24
Db4-C	1.15	25
Db6-C	1.63	24
Sym4-C	3.68	23
Sym6-C	5.85	26

In other words, one is assessing whether an additional latent variable improves the loss function (H_0 is rejected) in a statistically significant sense. In case the null hypothesis is rejected (*i.e.*, the model with $k + 1$ latent variables is better than the model with k latent variables, in a statistically significant sense), the algorithm continues and an additional latent variable is tested.

Otherwise, the algorithm stops, and the number of retained latent variables is set to the largest k for which the null hypothesis is rejected. In this work, the hypothesis test is applied with a significance level of 0.15 following the recommendations in [68].

Appendix B. Bayesian optimization for hyperparameter tuning

Bayesian optimization aims to solve the following optimization problem:

$$\max_{x \in \mathcal{X}} f(x) \quad (\text{B.1})$$

where $f(x)$ is an unknown and expensive function, with no available gradients or closed-form expressions. It is assumed that each objective function evaluation has zero mean Gaussian noise, $y = f(x) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma)$. BO relies on an probabilistic surrogate model of the objective function and an auxiliary acquisition function that balances exploration with exploitation to search for the global optimum of $f(x)$. As a surrogate model, we consider a zero mean Gaussian Process (GP), with covariance function, or kernel, $k(x, x')$ [69]:

$$f(x) \sim \mathcal{GP}(0, k(x, x')), \quad (\text{B.2})$$

where x and x' are vectors with dimensionality D . The integer valued hyperparameters are treated as continuous variables for the GP model

fitting and rounded to the nearest integer for evaluation of the objective function. Some of the CNN hyperparameters represent categorical variables, such as the type of activation function or pooling layer. Due to the mixed search space, with both continuous and categorical variables, we consider a separable kernel for the GP, such that the covariance between two sets of hyperparameters is given by:

$$k(x, x') = k_c(x_c, x'_c) \times k_r(x_r, x'_r) + k_c(x_c, x'_c) + k_r(x_r, x'_r) \quad (\text{B.3})$$

where k_c is a kernel based on Hamming distance for the categorical hyperparameters, x_c , and k_r is a kernel based on Euclidean distances, in this case a scaled Matérn 5/2 kernel, for the continuous hyperparameters, x_r . The latter are normalized to be in the range of $[0, 1]$ and the outputs y are standardized to be of zero mean and unit standard deviation. In addition, we use input warping [57] for all non-categorical hyperparameters, where each input is transformed using an individual parametric function, in this case a Beta CDF. Then, the scaled Matérn kernel is applied on the transformed input space. This specific function can provide a variety of different warpings and has found to perform well empirically for hyperparameter optimization tasks [57]. We place a hyper-prior on the GP hyperparameters (warping function, kernel and likelihood) and use Maximum A Posterior (MAP) estimation for inference of the GP model. For the likelihood and kernel, we use the default priors available in Ax/BoTorch while for the Beta CDF warping parameters, we consider the ones recommended by [57].

In the BO framework, an acquisition function that incorporates the surrogate model uncertainty is maximized to propose new experiments to search for the optimum hyperparameters. In this work we rely on the Noisy Expected Improvement (NEI) acquisition function [70], which is well suited for highly noisy settings, such as hyperparameter optimization of DL models. The logarithm version of this acquisition function is further considered, due to improved numerical performance [71]. The categorical hyperparameters lead to a mixed variable type optimization problem for the acquisition function maximization. Because there are only two categorical hyperparameters with a low number of choices, we enumerate all possible combinations and optimize the acquisition function with continuous variables for each category combination. The combination with the highest value is chosen as the next candidate for evaluation. The GP model fitting and acquisition function optimization

Table D.3
Hyperparameters of the CNN models for the WLO case study.

Hyperparameter	Raw	MC	MSC	SNV	SGD-2-9	SNV-SGD-2-9	Haar-R	Db6-R	Sym6-R
Lowest wav. scale retained	–	–	–	–	–	–	2	2	3
Hidden layers	3	2	3	2	3	2	3	2	2
Hidden units	82	53	64	73	71	97	10	69	20
Kernel size	184	152	247	72	140	256	48	48	130
Regularization weight	1×10^{-5}	0.0208	0.0470	0.0024	0.0012	0.0008	0.0325	1×10^{-5}	0.0010
Learning rate	0.0008	0.0031	0.0032	0.0044	0.0033	0.0065	0.0012	0.0014	0.0009
Pooling type	Max.	Avg.	Max.	Avg.	No	Max.	No	Max.	Max.
Activation function	Tanh	ELU	ELU	ELU	ELU	Tanh	Tanh	Tanh	Tanh

use the default optimizer in Ax/BoTorch. We consider 50 initial samples using a Sobol design and 150 total iterations. In addition, we use early stopping, where the BO algorithm stops if the current best observed value does not improve over 1% when compared with the inter-quartile range of all the observations.

Appendix C. Hyperparameters for the beer case study

See Tables C.1–C.3.

Appendix D. Hyperparameters for WLO case study

See Tables D.1–D.3.

Data availability

The code used in this work is openly available in https://github.com/PClabUC/model_preprocessing_comparison_spectral_analysis. Due to confidentiality reasons, only the data and code from the beer dataset can be published.

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